

2024*Time : 3 hours**Full Marks : 70*

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section-A**Objective type Questions****Compulsory**

1. Choose the correct option from the following multiple choice questions :

$$2 \times 10 = 20$$

(i) What is an operating system ?

- a) interface between the hardware and application programs
- b) collection of programs that manages hardware resources
- c) system service provider to the application programs
- d) all of the mentioned

(ii) In Operating Systems, which of the following is/are CPU scheduling algorithms?

- a) Priority
- b) Round Robin
- c) Shortest Job First
- d) All of the mentioned

(iii) The operating system is responsible for?

- a) bad-block recovery
- b) booting from disk

- c) disk initialization
 - d) all of the mentioned
- (iv) Whenever a process needs I/O to or from a disk, it issues a _____.
- a) system call to the operating system
 - b) a special procedure
 - c) system call to the CPU
 - d) all of the mentioned
- (v) In an operating system, each process has its own _____.
- a) address space and global variables
 - b) open files
 - c) pending alarms, signals, and signal handlers
 - d) all of the mentioned
- (vi) A process can be terminated due to _____.

- a) normal exit
- b) fatal error
- c) killed by another process
- d) all of the mentioned

(vii) What is interprocess communication?

- a) communication within the process
- b) communication between two process
- c) communication between two threads of same process
- d) none of the mentioned

(viii) The time taken to move the disk arm to the desired cylinder is called the

_____.

- a) positioning time
- b) random access time
- c) seek time
- d) rotational latency

- (ix) Physical memory is broken into fixed-sized blocks called _____.
- a) frames
 - b) pages
 - c) backing store
 - d) none of the mentioned
- (x) The _____ table contains the base address of each page in physical memory.
- a) process
 - b) memory
 - c) page
 - d) frame

Section-B

Short Answer type Questions

Answer any **four** questions of the following :

$$5 \times 4 = 20$$

2. Discuss Virtual Memory and its benefits.

3. What is FCFS? Explain briefly with a diagram.
4. Differentiate between Time Sharing and Real Time Operating Systems.
5. Define an Operating system and its services.
6. Explain Demand Paging.
7. Differentiate between pre-emptive and non-preemptive scheduling.
8. Discuss various allocation methods of a file.
9. Define Deadlock.

Section-C

Long Answer type Questions

Answer any **two** questions of the following :

$$15 \times 2 = 30$$

10. Consider the following page reference string
1,2,3,4,1,2,5,1,2,3,4,5.

How many page faults would occur for the following replacement algorithms, assuming four frames :

- a. LRU replacement
- b. FIFO replacement
- c. Optimal replacement

11. Consider the following process :

Process	Arrival time	Burst time
P1	0	8
P2	1	4
P3	2	9
P4	3	5

Calculate the average waiting time, turnaround time by SJF, FCFS and round robin algorithm.

12. Write short notes on :

- a. FIFO page replacement
- b. Multitasking Operating System
- c. Thread

13. Describe Paging and Segmentation techniques of memory management in details.

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